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Chaos in three-body dynamics: Kolmogorov–Sinai entropy

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ABSTRACT

An ensemble of Newtonian three-body models with close initial separations is investigated by following the evolution of a 'drop' in the homology map. The onset of chaos is revealed by the motion and the complex temporal deformation of the drop. In the state of advanced chaos, the drop spreads over almost the whole homology map, quite independently of its initial position on the map. A general quantitative measure of this process is the mean exponential rate of spreading, which bears resemblance to Kolmogorov–Sinai entropy; this is introduced and estimated in terms of the homology mapping. In a similar manner we also estimate the mean exponential rate of divergence of initially close-by trajectories. This is a close analogue to the Lyapunov exponent. These parameters measure two complementary aspects of dynamical instability, which is the basic mechanism of the onset of chaos.

Key words: chaos – instabilities.

1 INTRODUCTION

The concept of entropy, introduced initially for statistical systems with a large number of degrees of freedom (such as a fluid), was extended later to dynamical systems which exhibit stochastic behaviour (Kolmogorov 1958, 1959; Sinai 1959). In the latter case, the number of degrees of freedom can be small. Arnold (1978) demonstrates that dynamical chaos can appear when the number of degrees of freedom is at least two. To study the onset of chaos in three-body dynamics, we introduce in the present paper a quantity which bears a resemblance to the Kolmogorov–Sinai entropy.

In three-body systems, the non-linear interactions of the gravitating bodies are thought to be the cause of local dynamical instability, which is the basic physical mechanism of chaotic behaviour of the systems (Szebehely 1972). The high sensitivity to fine details of the initial conditions is the major manifestation of dynamical instability. It is characterized quantitatively by the Lyapunov exponent.

We will show here that the concept of Kolmogorov–Sinai entropy enables us to clarify an aspect of the process that is complementary to (while not independent of) those described by the Lyapunov exponent. A close inter-relation between these two fundamental characteristics of chaos is demonstrated below within the context of a gravitational three-body system.

We introduce and estimate the Kolmogorov–Sinai entropy for three-body dynamics in terms of the homology mapping. This method (Agekian & Anosova 1967) has proven to be an effective tool for both the analysis and the visualization of the results (Ivanov & Chernin 1991; Chernin & Valtonen 1998).

In our previous paper (Heinämäki et al. 1998) we introduced the concept of intermittency to three-body dynamics, using homology mapping to characterize alternations of apparent order and disorder that occur in an unpredictable manner during the lifetime of a three-body system.

2 KOLMOGOROV–SINAI ENTROPY: A PHYSICAL DEFINITION

An exact formal definition of Kolmogorov–Sinai entropy is quite complicated in terms of mathematics, and we will not present it here. We give instead a physical description which is rather simpler, and adequate to a reasonable accuracy (Zaslavsky 1985; Sagdeev, Usikov & Zaslavsky 1988; Zaslavsky et al. 1991). From here we will for the sake of brevity call our adopted physical description K–S entropy, although it does not strictly fulfil all the mathematical requirements of the definition.

Consider a 'phase drop' which is an ensemble of all possible states of the system within a finite volume of the phase space of the system. In Hamiltonian systems, the 'liquid' of the drop is incompressible, and the density of the drop elements is constant during the whole evolution. This means that the total volume of the drop $\Delta\Gamma$ is also constant, and the spreading of the drop over the phase space is completely as a result of the deformations in its shape. Typically, multiple branches emerge from the drop liquid, which get thinner and longer with time. This is the way in which the drop can fill various regions of the phase space.

Looking at the evolution of the drop with a certain finite resolution $\varepsilon \leq \Delta\Gamma$, one cannot see all the fine details of its shape, but only larger forms that are above the resolution limits. Below these limits, the unresolvable details reveal a smooth distribution of the phase liquid. The density of this smoothed distribution is obviously less than the actual (constant) density of the liquid.

Let us define a phase-space volume $\overline{\Delta\Gamma}$ by measuring a convex

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hull containing all the points of our drop. The 'coarsened liquid' within the volume $\overline{\Delta\Gamma}$ is compressible. The local density varies within $\overline{\Delta\Gamma}$. Note that ratio $\Delta\Gamma/\overline{\Delta\Gamma}$ can be understood as a measure of the filling factor of the evolved drop.

The volume $\overline{\Delta\Gamma}$ grows with time, and this growth is typically exponential in the systems that are able to develop chaotic behaviour: $\overline{\Delta\Gamma}(t) = \Delta\Gamma \exp(ht)$. Here h is called Kolmogorov–Sinai (K–S) entropy.

According to the 'standard' definition of entropy in statistical physics, a phase volume $\Delta\Gamma$ is characterized by entropy which is equal to

$$S = \ln(\Delta\Gamma).$$

In the case of dynamical systems (Zaslavsky 1985),

$$\overline{S} = \ln(\overline{\Delta\Gamma} = ht + \ln(\Delta\Gamma).$$

So K–S entropy h is not entropy in a 'standard' sense, but a growth rate of the 'coarsened drop'.

In a fine-grain description, the same process can be characterized qualitatively by the largest postitive Lyapunov exponent.

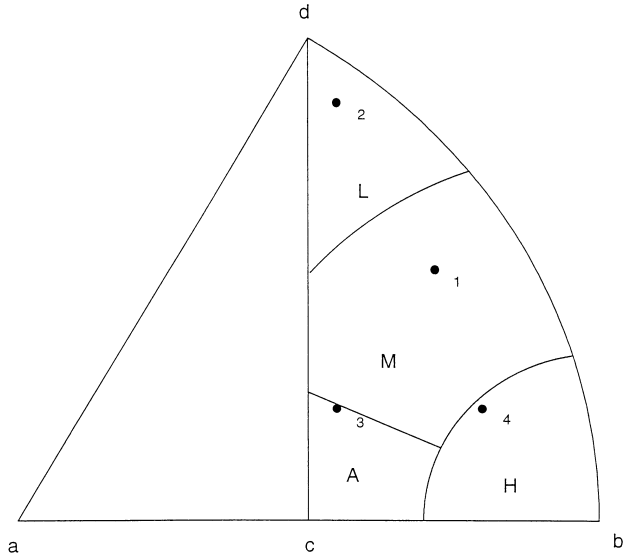


Figure 1. The homology map, after Agekian & Anosova (1967). Any triangle is represented by a point located in accordance with its shape, independently of its proper size. In this map one rescales the triangle such that the longest side becomes a constant interval ab and it meets the shortest side at point b. The centre of the longest side is denoted by c: ac = cb, while point d is defined as the crossing point of the arc of radius ab with the centre at a and the normal cd to ab. Thus any triangle can be plotted as a point in the 'curved triangle' cdb and the location of the point corresponds to the homologous structure of the triangle. It is easy to see that a triangle with nearly equal sides is represented by a point in the upper corner of the map; a triangle with one side much smaller than the two others finds its place in the right bottom corner of the map; a triangle with its three corners lying almost along a straight line is represented by a point located near the bottom of the map. In accordance with these considerations, one can define four characteristic areas in the map: the top corner of the map may be called the area of Lagrangian (L) triangles – after the well-known solution of the classical three-body problem; the right bottom corner of the map, where the triplets with close binaries are located, may be referred to as the area of hierarchical (H) configurations; the zone near the bottom of the map may be defined as the area of aligned (A) configurations; and the remaining area is found in the middle (M) of the diagram. The initial positions of four 'homology drops' (see the text) are indicated.

These quantities are not completely independent of each other and in general cannot be reduced to each other, except in a special case specified by Pesin (1977) and Lichtenberg & Lieberman (1992).

3 SIMULATIONS OF THE HOMOLOGY DROP

The concept of the phase drop may be reformulated and applied to a three-body system in a natural way by the use of homology mapping (Fig. 1). The homology map may be considered as a two-dimensional scale-invariant hypersurface of the phase space of the three-body system. Its geometry is Euclidean and the whole area is finite. The homology mapping is an elegant way of analysing the evolution of the system and – which is also important – visualizing the results of the analysis. One can imagine a set of all possible homology states within a small region of the homology map and consider it as a 'homology drop', in analogy with the phase drop. Following the evolution of the homology drop, one can answer two questions. How strong are the deformations of the drop with time? How fast do they occur?

In non-linear physics it has been shown (e.g. Schuster 1982) that a low-dimensional section can represent adequately the dynamical structure of an evolving system. Analogously, a drop moving on the two-dimensional hypersurface of the phase space will reflect the general features of the behaviour of the system in the $6 \times N -$ dimensional phase space. Because our simulations are restricted to a common plane, the complete behaviour would be described in a $4 \times N -$ dimensional phase space.

We simulated four homology drops and followed their evolution. Each drop contained 100 separate initial states, which

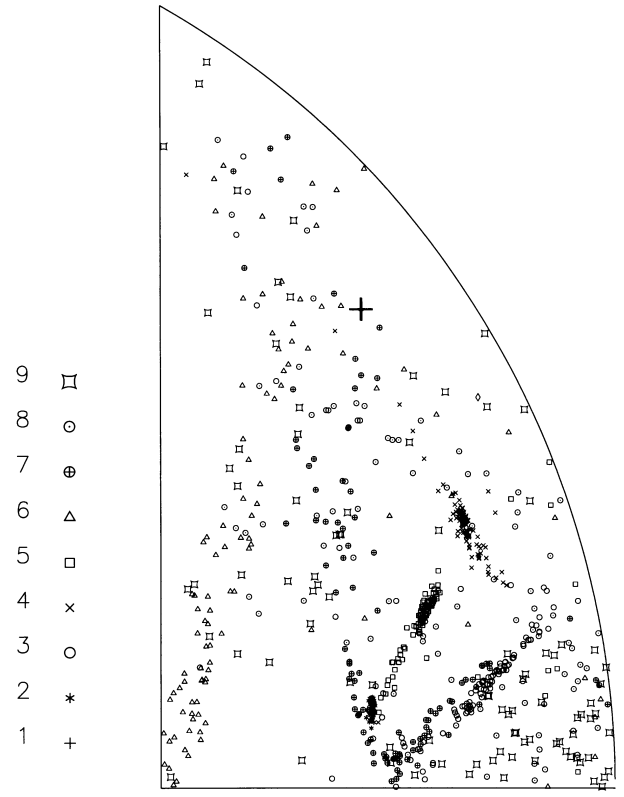


Figure 2. The moving homology drop: the evolution of an ensemble of 100 three-body systems starting from nearly identical initial conditions in the M-area of the homology map; the state of the ensemble is demonstrated in nine subsequent time moments: $\tau = 0$ (No. 1), $\tau = 1.75$ (No. 2), $\tau = 3.5$ (No. 3), …, $\tau = 14$ (No. 9).

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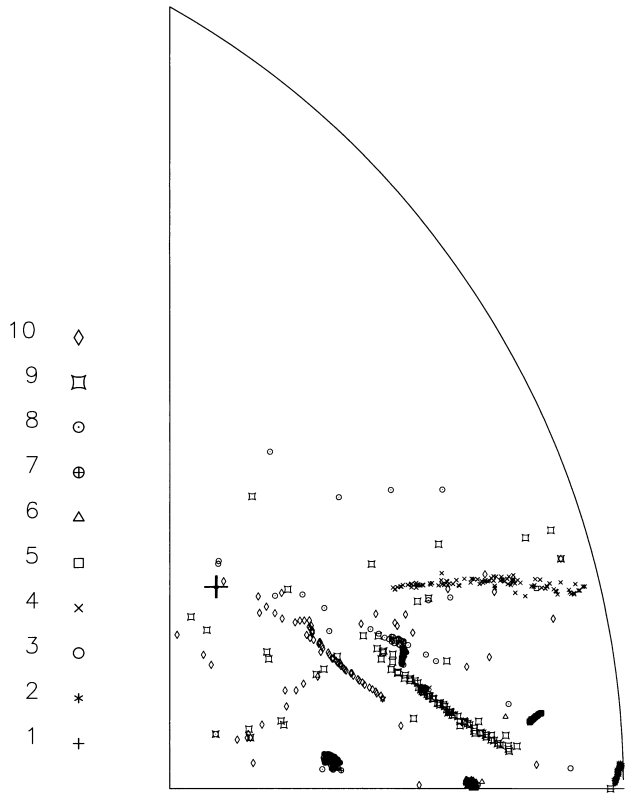


Figure 3. As Fig. 2, but with the homology drop initially in the A-area of the homology map. 10 subsequent time moments are shown: $\tau = 0$ (No. 1), $\tau = 1.75$ (No. 2), $\tau = 2.7$ (No. 3) and then in steps of 0.9τ.

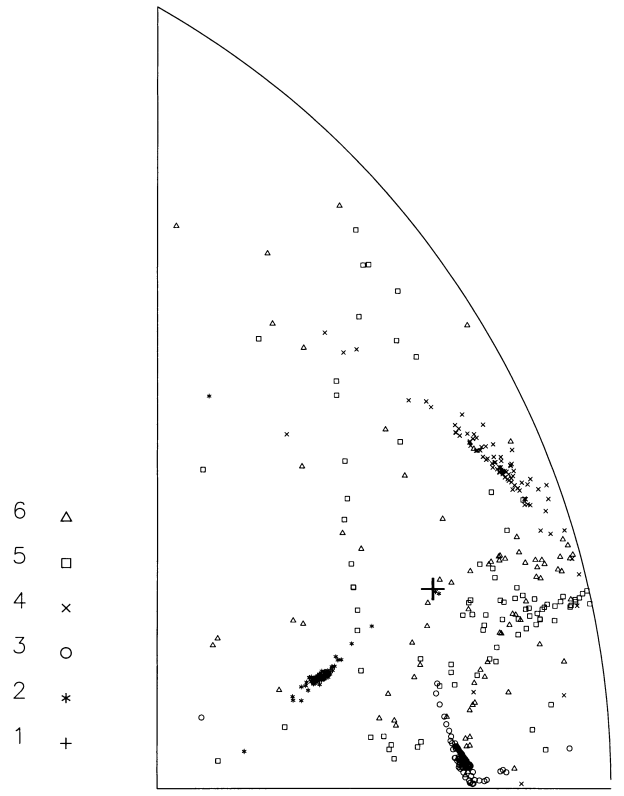


Figure 4. As Fig. 2 but with the homology drop initially in the H-area of the homology map. Six subsequent time moments are shown: $\tau = 0$ (No. 1), $\tau = 1.0$ (No. 2), $\tau = 1.75$ (No. 3), $\tau = 2.5$ (No. 4), $\tau = 3.25$ (No. 5) and $\tau = 4.0$ (No. 6).

were spread along a horizontal linear segment with a total length of 10^{-3} in the homology map. This means that neighbouring triple-body systems within each drop were identical except that the body opposite to the longest separation ab $\equiv 1$ was moved by 10^{-5} in the direction of ab (Fig. 1). The 100 states (or triple-body systems) do not definitely constitute 'all possible states' within the small volume, but they represent an adequate enough sample to recognize the major features of the evolution and motion of the homology drop (Figs 2–4).

The four drops were initially located in the areas M, L, H and A of the homology map. The three bodies were assumed to have equal masses and initially zero velocities. The dynamical evolution of each drop was followed until all the systems experienced an escape. An escape occurs when a single body in the triple system acquires positive total energy.

The evolution of the drop with an initial position in the M-area of the homology map is presented in Fig. 2. Eight subsequent stages of the evolution are demonstrated in the moments of time $t = 0, 1.75\tau_0, 3.5\tau_0, 5.25\tau_0, \ldots, 14\tau_0$ measured in terms of the 'initial crossing time' $[\tau_0 = (r_0^3/3Gm)^{1/2}$, where G is the gravitational constant, r_0 the longest distance in the system, and m is the total mass of the system]. All references to time from here on are made in units of τ_0. Within the first crossing time, the ensemble as a whole moves from the initial M-area (marked with a large cross) in the map to the bottom of the map. The drop spreads and deforms, but the ensemble preserves its coherence almost completely. By $\tau = 3.5$ (symbol number 3) the drop has moved to the H-area and takes the form of a stripe overall, while several members have already been scattered to other areas of the map. At the next stage of the evolution ($\tau = 5.25$, symbol number 4), the drop moves to the boundary of the H-area with a further deformation of the stripe. At $\tau = 7$ (symbol number 5), the stripe turns again to the H-area; several earlier separated members of the ensemble come to this area too. At $\tau = 8.75$ (symbol number 6), the drop breaks up and all the members spread over almost the entire homology map. A new trend in the evolution starts at this point. More and more members visit the H-area of the map, effectively reducing the spread of the points over the homology map. As a triple system breaks up it will finally migrate to the H-area.

A more or less similar evolution with almost the same transient states is observed in homology drops starting their evolution in the A- and H-areas, as seen from Figs 3 and 4. The evolution of the drop with initial position in the L-area of the map (Fig. 5), however, is quite different. This area contains the states with strong three-body interplay, which is obvious from the fact that the separations between the bodies are comparable here. The states of interplay are highly unstable. The lifetime of the homology drop is very short here, with immediate evolution to the H-area (Fig. 6) and the decay of all 100 systems after the first interaction at $\tau \approx 1$.

4 COARSE EVOLUTION OF THE DROP AND KOLMOGOROV–SINAI ENTROPY

To quantify the process of development of chaos, we introduce a measure of spreading (or scattering) of the drop in the homology map analogous to the coarse-grained description in statistical mechanics (Zaslavsky 1985). We define a fractional area of the map, which contains all the members of the ensemble: $S = A/A_0$,

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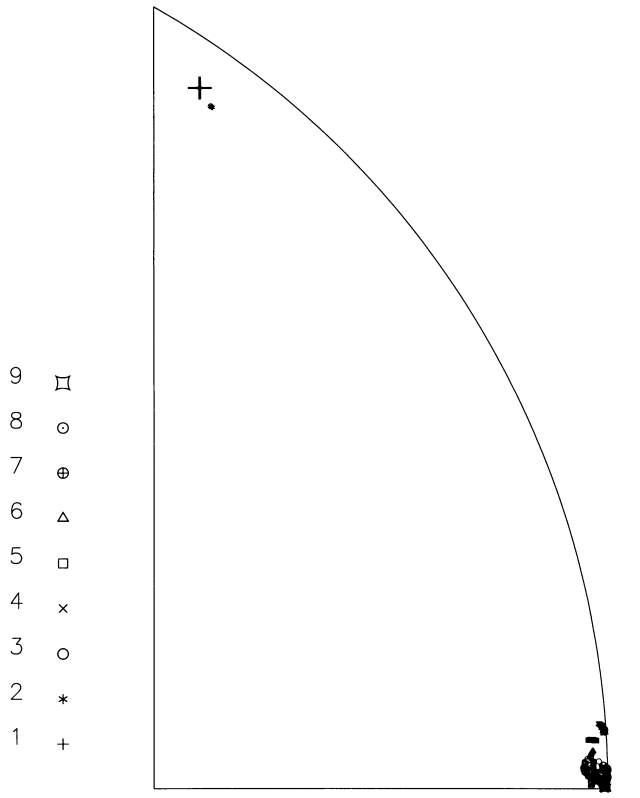


Figure 5. As Fig. 2 but with the homology drop initially in the L-area of the homology map. Nine subsequent time moments are illustrated: $\tau = 0$ (No. 1), $\tau = 0.5$ (No. 2), $\tau = 1.5$ (No. 3), $\tau = 2.5$ (No. 4), $\tau = 3.5$ (No. 5), …, $\tau = 7.5$ (No. 9).

where A is the area of the convex hull containing all the members of the drop and A_0 is the total area of the map. Note that the area $A(t)$ includes not only the representative points, but also the empty space between the points. $S(t)$ or $A(t)$ have here the same physical meaning as $\overline{\Delta\Gamma}$ of Section 2.

The value of S as a function of time is presented in Figs 7–9 for the drops with initial positions in the M-, A- and H-areas of the homology map, respectively. We see that S behaves rather erratically. However, three different stages can clearly be recognized in the overall evolution. The initial stage is characterized by a rapid increase of S superposed on strong oscillations. Then $S(t)$ reaches its maximal value ~ 1, and stays at an almost constant level with only weak oscillations. Finally when more and more escapes occur from the systems, $S(t)$ decreases rapidly.

For comparison, Fig. 10 shows the number of systems in the ensemble as function of time.

At the stage of rapid evolution, the growth of $S(t)$ is nearly exponential: $S \propto \exp(ht)$. The average growth rate $h > 0$ has a close analogue in the K–S entropy, defined in a similar manner for the whole phase space (see Section 2). The physical meaning of K–S entropy introduced in this way to three-body dynamics is obvious: the greater the quantity h, the more rapid is chaotic spreading of the ensemble over the homology map, and therefore the stronger is the ability of the system to develop chaotic motions in this way. The value of h is given in Tables 1 and 2 for the three homology drops. Two types of time interval are used in the calculations: the first one is from the beginning of the evolution to the moment of the first escape from the system (Table 1); and the second one is from $\tau = 1$ to the moment when escapes have occurred in 5 per cent of the systems. The values of h are given in units of the inverse crossing time.

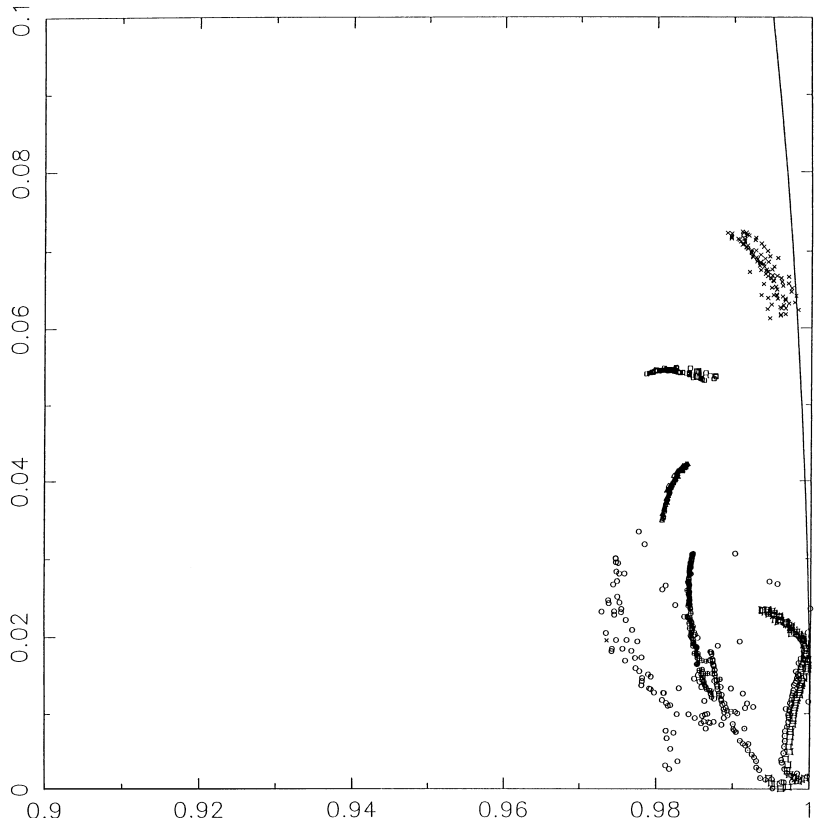


Figure 6. As Fig. 5, with a magnified H-area.

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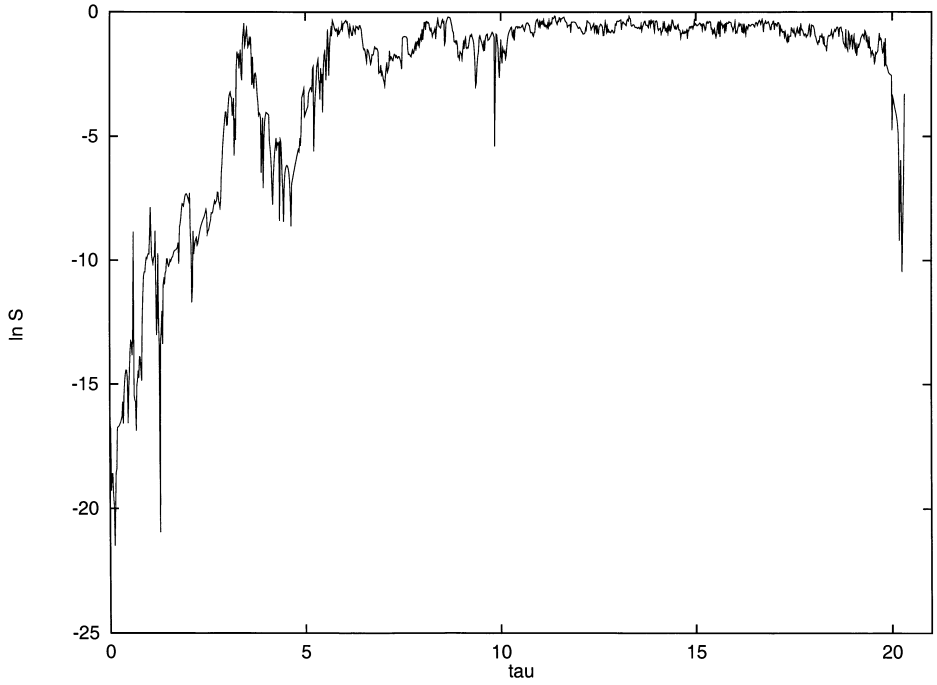


Figure 7. The evolution of areas of the drop $S = A(t)/A_0$, for the drop initially in the M-area of the homology map.

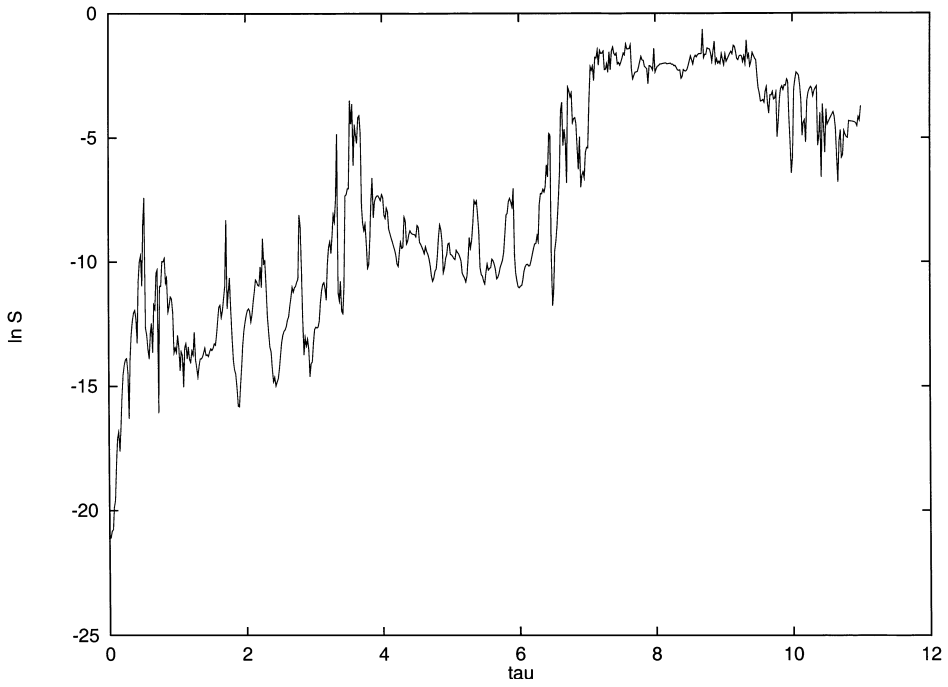


Figure 8. As Fig. 7, but for the A-area of the homology map.

From Tables 1 and 2 we can see that by the first definition h is larger (Table 1) than by the second one (Table 2). In both cases, however, K–S entropy lies in an interval: $h \approx (0.6\text{–}6)\tau^{-1}$. The median value, $h \approx 1\tau^{-1}$, may be considered as a characteristic value of h in three-body dynamics.

Note that the initial growth of $S(t)$ in Figs 7–9 is driven by large bursts that are generated by the process of exchange of the binary members in a hierarchical state of the system (as is seen from our simulations).

5 FINE STRUCTURE OF THE MOVING DROP AND THE LYAPUNOV EXPONENT

The 'fine' structure of the homology drop reveals the behaviour of individual trajectories within the homology mapping of the drop. This is in contrast to the coarse-grained description which follows the shape of the drop as a whole.

Fig. 11 shows the evolution of the trajectories of 100 systems within the homology drop located initially in the M-area. The

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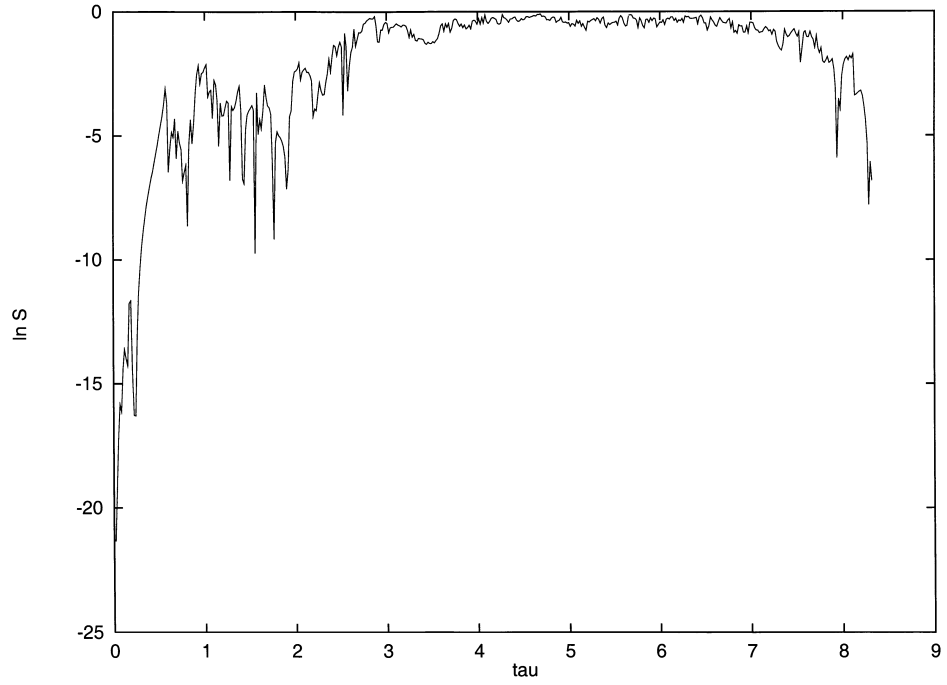


Figure 9. As Fig. 7, but for the H-area of the homology map.

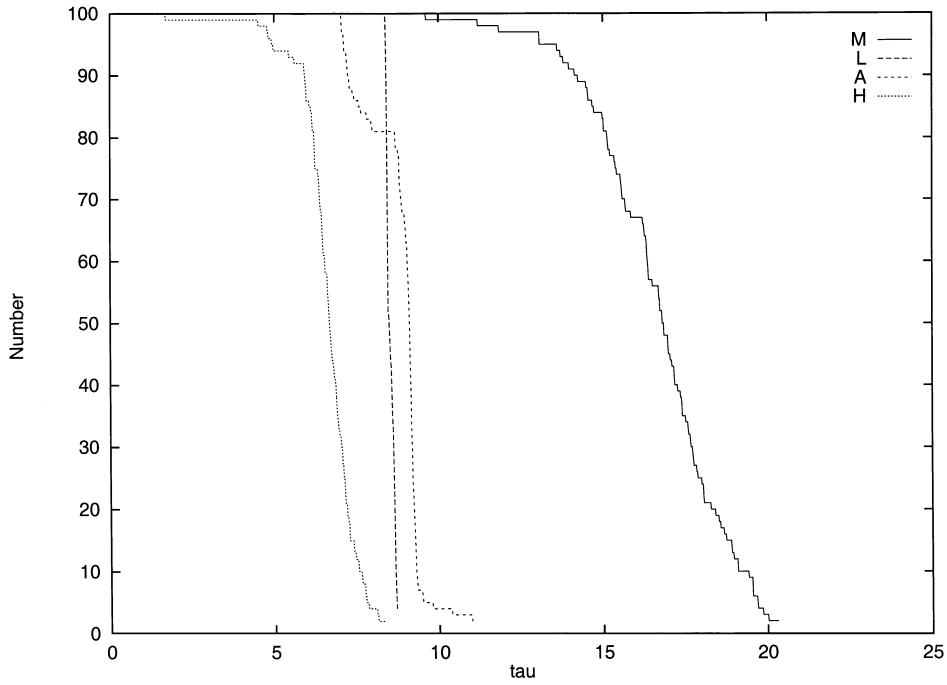


Figure 10. The number of members in each ensemble as a function of time.

three sides of the homology triangle (which are the three separations between the bodies), $d_i, i = 1, 2, 3$, are shown in the figure as a function of time. Each side of the initial triangle is represented by a different colour: blue the distance between bodies located in corners a and b; green between a and body labelled with a number; red between b and body labelled with a number. If the system has a close binary, then one curve oscillates at small d while the two other curves oscillate at high d. A change of a member in the binary system is obviously shown as a change in colour of the lines representing the binary pair.

The triangles are initially very similar to each other in geometry, so their differences are not observable in Fig. 11 at $\tau = 0$. After the first close encounter at $\tau \approx 1$ the visible differences appear in the plotted orbits. These differences increase with time, which is revealed by the widening of the initially narrow curves into broader stripes. Qualitatively, the evolution of a typical system proceeds (Fig. 11) through steps that are well-known in the three-body problem: formation of a temporary binary, exchange of the binary members, close three-body interactions which alternate in a chaotic manner, and finally escape of a member from the system. In the ensemble of 100 systems which make up the homology drop, all of this leads to a

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chaotic divergence of the system trajectories, obvious from Figs 7–9.

Figs 12 and 13 show the evolution of the homology drops located initially in the A- and H-areas: these pictures are qualitatively rather similar to Fig. 11, but show different divergence rates of trajectories. The stage of rapid development of chaos starts after an initial phase of rather slow evolution; this phase is shorter for the H-case and it is longer for the A-case.

Fig. 14 shows the evolution of the trajectories when the homology drop is located initially in the L-area. In this case, the decay of all of the systems occurs after the first close three-body interaction. The lifetimes of these systems, in contrast to the previous case, are not long enough for the development of dynamical chaos.

The stage of rapid development of chaos is when the most rapid divergence of individual trajectories occurs. The distance between two initially close trajectories reveals exponential growth $\delta_i \propto \exp(\sigma_i t), i = 1, 2, \ldots, 100$ (Figs 11–14). Here δ_i is the distance between two trajectories which is presented as the width of the stripe i formed by the same side (or colour) of the triangles in Figs 11–14. The growth rates σ_i have here the same physical meaning as the Lyapunov exponent σ in Section 2, defined for the trajectories in the phase space. Figs 15–17 show examples of various σ_i as slopes of the curves $\sigma_i(t)$, at the stage of most rapid divergence of trajectories.

Table 1. Kolmogorov–Sinai entropy h, the Lyapunov exponent $\langle\sigma\rangle$ and the ratio $h/\langle\sigma\rangle$ of the cases M, A and H are shown. The respective time epochs Δt and correlation coefficients r of the fits are indicated. The time interval used is from the beginning of the evolution to the moment of the first escape from the system.

	M 1	A 3	H 4
h	1.46	1.07	6.26
r	0.85	0.71	0.72
$\langle\sigma\rangle$	0.67	0.46	2.79
r	0.91	0.63	0.89
Δt	0–9.6	0–7.1	0–1.7
$h/\langle\sigma\rangle$	2.18	2.33	2.24

Table 2. Kolmogorov–Sinai entropy h, the Lyapunov exponent $\langle\sigma\rangle$ and the ratio $h/\langle\sigma\rangle$ of the cases M, A and H are shown. The respective time epochs Δt and correlation coefficients r of the fits are indicated. The time interval used is from $\tau = 1$ to the moment when escapes have occurred in 5 per cent of the systems.

	M 1	A 3	H 4
h	0.73	1.09	1.27
r	0.79	0.70	0.79
$\langle\sigma\rangle$	0.49	0.48	1.1
r	0.91	0.59	0.90
Δt	1.1–13.1	0.7–7.1	0.95–4.9
$h/\langle\sigma\rangle$	1.49	2.27	1.15

In Figs 15–17, the mean values of $\langle\sigma\rangle$ are also shown (solid lines) as averaged over all the trajectories of three homology drops. For the averaged values, an exponential growth is also observed at the stage of the most rapid development of chaos: $\langle\delta\rangle \propto \exp(\langle\sigma\rangle t)$. Here $\langle\sigma\rangle$ is the average Lyapunov exponent defined in this way, not for individual trajectories but for the ensembles that constitute the homology drops. The value of $\langle\sigma\rangle$ is presented in Tables 1 and 2 for the three examples of the drop and for two different time epochs.

Figs 15–17 show another way of averaging σ (upper curves). This deals with the total width of the stripe formed by all 100 trajectories together. The slope of these three curves $\bar{\sigma}$ is close to what is observed in $\langle\sigma\rangle$ curves for the drops with starting location in the areas A and M of the homology map; for the drop moving from the H-area $\bar{\sigma}$ differs from $\langle\sigma\rangle$. The latter case exhibits the most violent chaotic behaviour.

We see from Tables 1 and 2 that the values of $\langle\sigma\rangle$ are in the interval $\langle\sigma\rangle \approx (0.5\text{–}3)\tau^{-1}$, with the median at $\langle\sigma\rangle \approx 0.5/\tau$. To order of magnitude, this characteristic value is close to that found for K–S entropy (Section 4.): $\langle\sigma\rangle \approx h$.

6 DISCUSSION

We have considered three-body systems in the manner in which physicists consider K-systems. We have introduced the concept of K–S entropy adapted to the physics of three-body systems, and shown that these systems are characterized by $h > 0$.

We have also found that, to order of magnitude, K–S entropy is close to the Lyapunov exponent: $h \approx \sigma \approx \langle\sigma\rangle$. We can add now that, according to the previous studies of three-body chaos (Ivanov, Filistov & Chernin 1995), the inverse correlation decay time τ_c has the same order of magnitude. The relation $h \sim \sigma \sim 1/\tau_c$ is the major physical characteristic of systems with local dynamical instability (Zaslavsky 1985; Sagdeev et al. 1988; Zaslavsky et al. 1991; Ott 1993) (see also Section 2). It is obvious, however, that three-body systems have some properties that make them different from the Hamiltonian or dissipative dynamical systems usually considered in physics. First of all, almost all three-body systems have a limited lifetime in a wide range of parameters of the systems. They decay sooner or later because one of the bodies eventually escapes from the system. There are no asymptotic states in three-body dynamics towards which the system can evolve (but never quite reach) over an infinite length of time. Their entire life is an evolution that consists of sequences of intermittent transient states and ends abruptly with the final escape of a system member. The escape means that one of the configuration coordinates of the phase space is infinite, and so the phase space of the system is not compact. This fact lies in the very nature of N-body gravitating systems.

Because of this, most of the strong and rigorous general results related to systems with compact phase space in non-linear physics cannot be translated directly to the dynamics of three-body gravitating systems or N-body systems. Nevertheless, the general ideas and concepts of non-linear physics are actually more universal and can in a natural way be applied to a description of three-body chaos, taking into account the specific properties of few-body gravitating systems.

We have demonstrated the validity of the concept of K–S entropy for three-body dynamics by using homology mapping. The method of phase drop, re-formulated for homology mapping, has enabled us to calculate the value of K–S entropy h for a number of characteristic examples. In a similar way, we have

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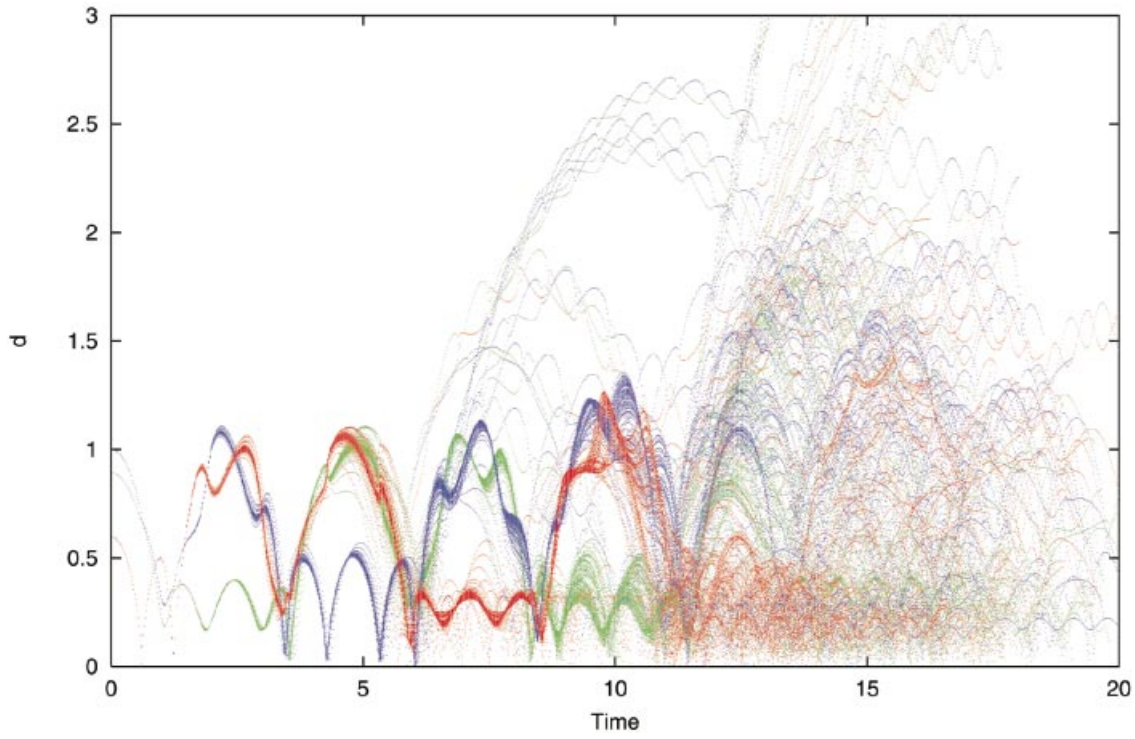


Figure 11. The divergence of trajectories in the homology drop with the initial position in the M-area of the homology map (time is in units of initial crossing times).

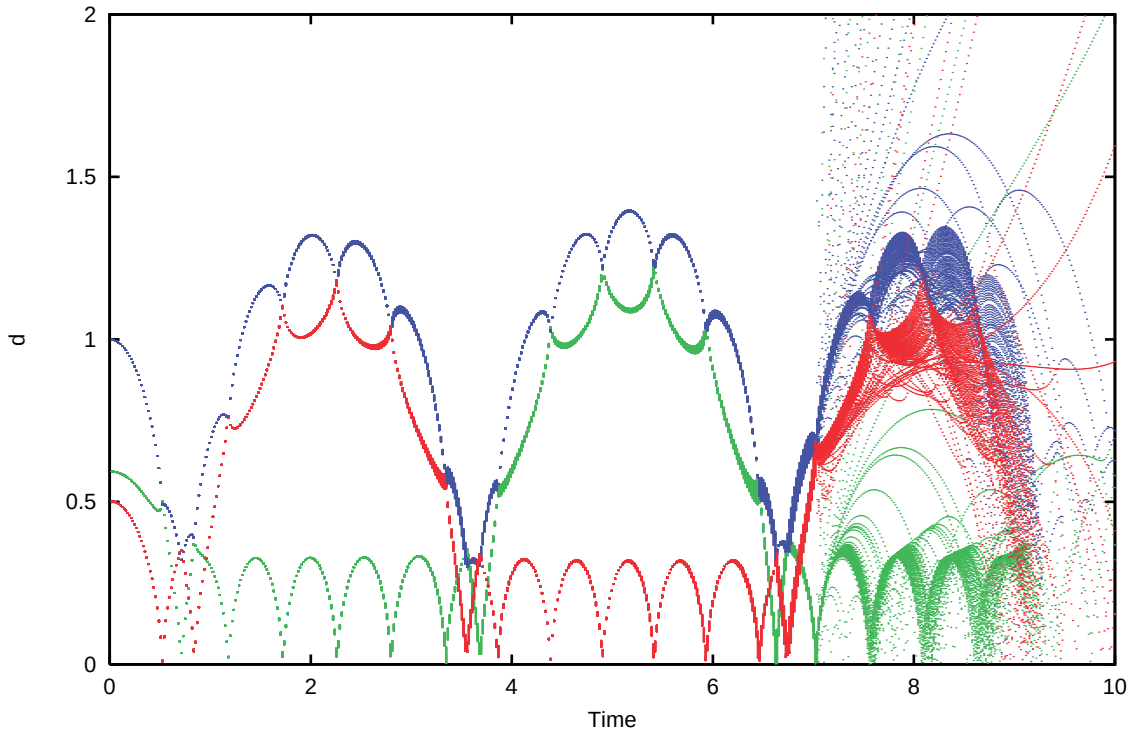


Figure 12. As Fig. 11 for the homology drop with the initial position in the A-area of the homology map.

found an analogy with the Lyapunov exponent which is a measure of the divergence of individual trajectories in the homology map. K–S entropy and the Lyapunov exponent correspond to coarse-grained and fine-grained statistical descriptions of the three-body chaos, respectively.

Three-body analogues of K–S entropy and the Lyapunov major exponent quantify two basic aspects of dynamical instability, as the key mechanisms of development of chaos. According to general ideas in non-linear physics, a system with chaotic behaviour has two complementary properties: (i) it is highly

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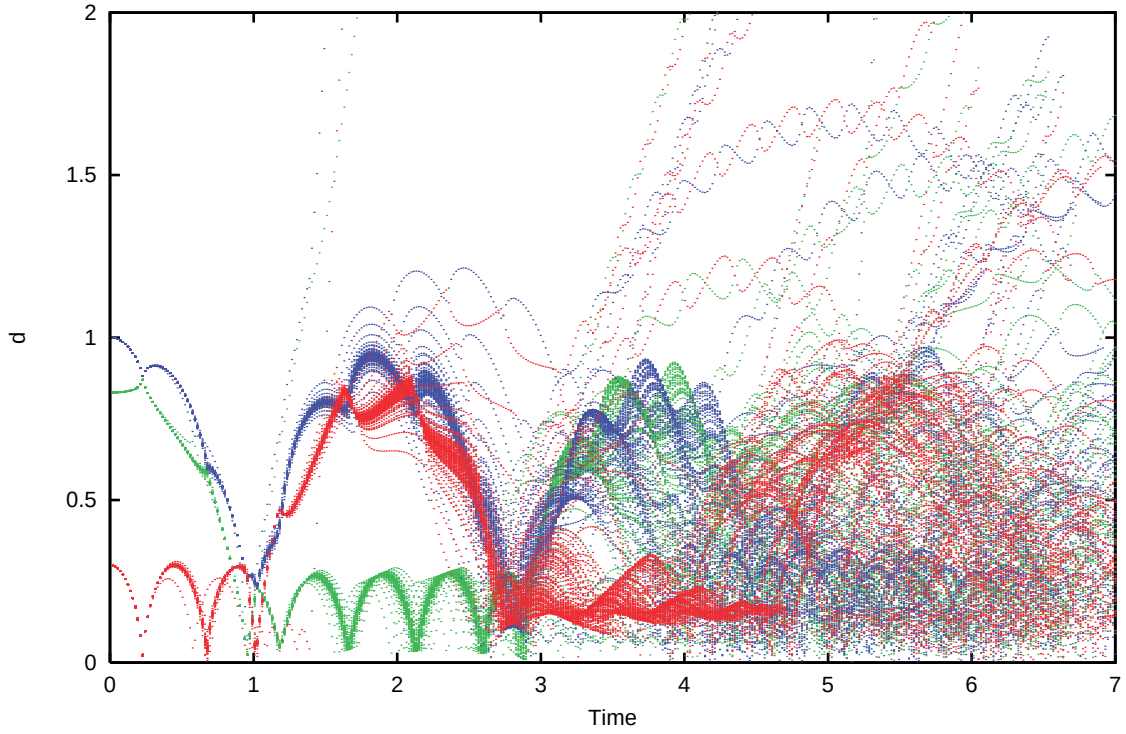


Figure 13. As Fig. 11 for the homology drop with the initial position in the H-area of the homology map.

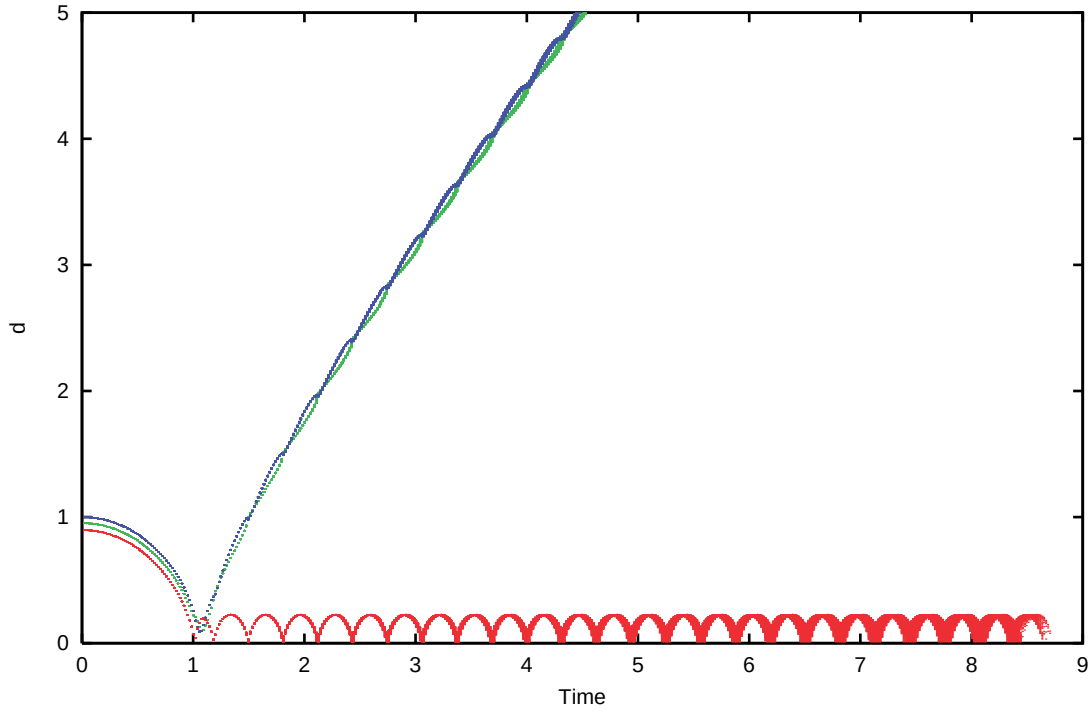


Figure 14. As Fig. 11 for the homology drop with the initial position in the L-area of the homology map.

sensitive to initial conditions; and (ii) it will forget the initial conditions with time. The sensitivity to initial conditions is mainly described by the Lyapunov exponent σ, because it measures the rate of individualization and divergence of the initially close trajectories. The ability to forget initial conditions is mainly estimated by K–S entropy h, because it shows how fast and to what extent the ensembles that start their evolution from different initial positions in the homology map can spread and occupy with time the entire manifold of all possible homological (and also dynamical) states.

We can observe, by visualizing the evolution of the moving homology drops, how these two basic tendencies interact in the

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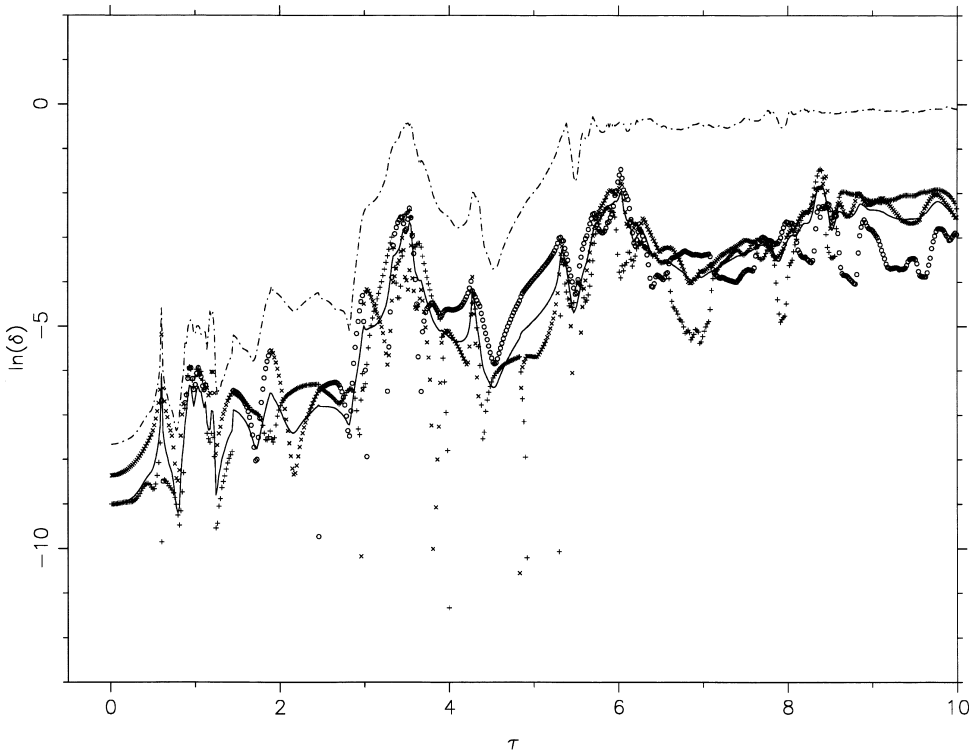


Figure 15. The divergence of nearby orbits for drops initially in the M-area of the homology map.

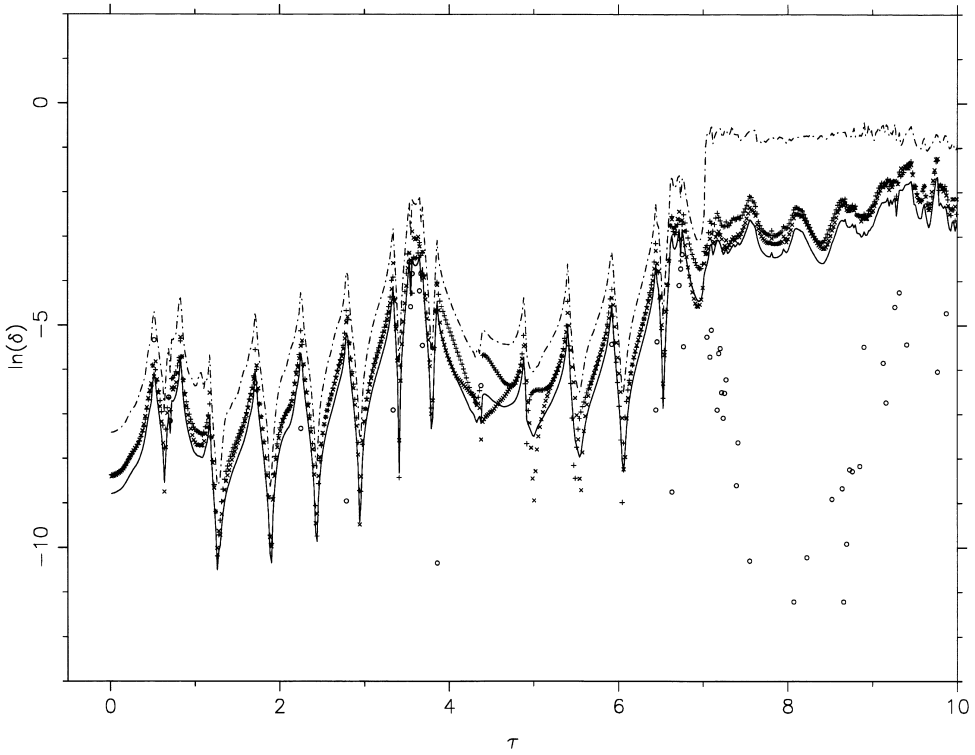


Figure 16. As Fig. 15, for the A-area of the homology map.

course of the onset of chaos. We find that this interplay, reflected in the values of h, σ and $\langle\sigma\rangle$, is rather complex. It characterizes the intrinsic complexity of the non-linear three-body dynamics. These quantities are obviously not independent because they describe the same process, but they do it in different ways. As a result, the relation between $\langle\sigma\rangle$ and h proves to be not of one-to-one type all the time. In particular, the largest $\langle\sigma\rangle$ for an ensemble does not mean that h is also the largest for the ensemble, as is seen in

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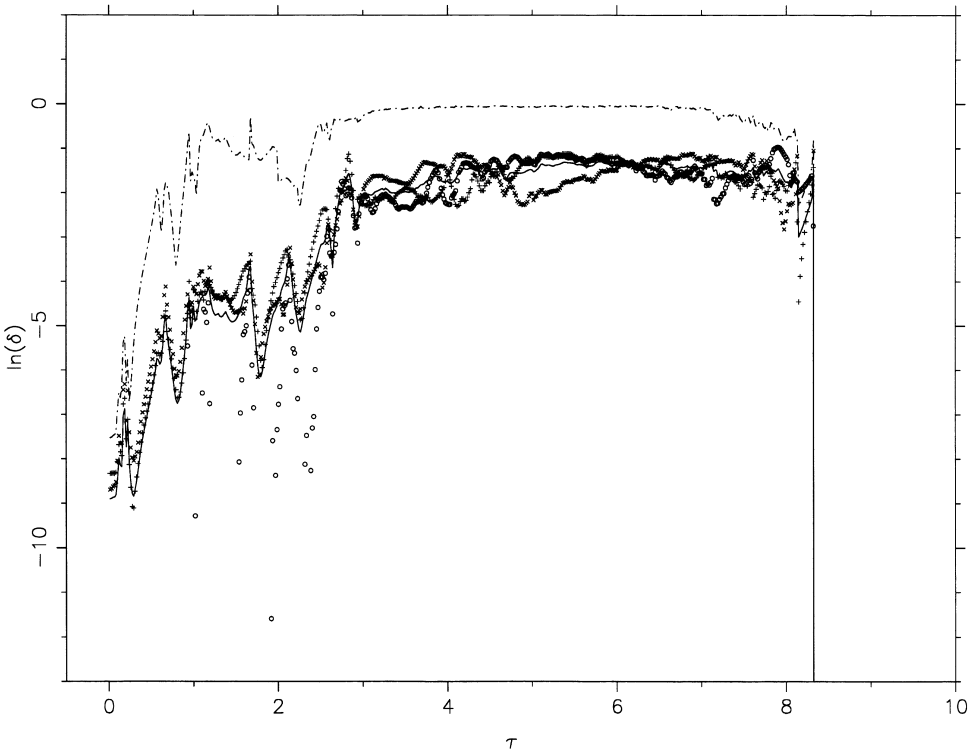


Figure 17. As Fig. 15, for the H-area of the homology map.

Table 2. In all the examples of Tables 1 and 2, the quantity h is found to be larger than $\langle\sigma\rangle$.

It is important that both tendencies cannot develop completely during the finite lifetime of the system. This is obvious from the fact that the quantities h and $\langle\sigma\rangle$ in Tables 1 and 2 are different for different initial positions of the homology drop. So the initial conditions have not been forgotten even when the homology drop has spread over almost all of the homology map. Systems that have small h appear to show more regular oscillatory behaviour.

If the structure of the homology map were uniform and homogeneous, we would expect the growth rate h derived from a two-dimensional measure and the growth rate σ derived from a one-dimensional measure to have a ratio of exactly 2. Tables 1 and 2 show that the actual ratio varies between values $h/\langle\sigma\rangle \simeq 1.2$ and 2.3. We feel that these differences are too large to be explained by errors. A natural explanation is as follows. If the homology drop remains string like in shape one should expect the ratio to be less than 2. If the spreading of the boundary of the homology drop can be characterized by formation of long fingers and deep cavities in a fractal manner, then ratios higher than 2 are expected. Both of these evolutions reflect clearly chaotic behaviour, with the latter being stronger. The latter systems are more prone to forget their initial conditions, while the former systems show more sensitivity to initial conditions.

On the other hand, the structure of the evolving drop is not uniform or homogeneous, even in the coarsened description. We can see from Figs 7–9 (and also Figs 11–13) that the representative points in the homology drop reveal two types of behaviour: a small fraction of them (10–20 per cent) spread fairly rapidly over almost all the area of the homology map, while the majority of the points (90–80 per cent) keep a certain degree of coherence for all the time of evolution. This means that a significant number of members of the ensemble do not have enough time to display a complete sensitivity to fine details of the initial conditions. In the calculations of Section 4, the estimates of h are dominated by the 'active' minority, and the estimates of $\langle\sigma\rangle$ are mostly determined by the 'conservative' majority. This may explain why K–S entropy is systematically greater than the average Lyapunov exponent in our experiment.

Note that the concept of K–S entropy, as well as the concept of intermittency and strange attractors, can be extended in a similar way using homology mapping to the relativistic three-body problem, e.g. the three-black-hole problem (Valtonen 1988, 1996; Valtonen & Mikkola 1991; Valtonen et al. 1994). Specific properties of dissipative chaos in three-black-hole systems, which are due mainly to energy and momentum losses via gravitational radiation, will be considered in a separate work.

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